

RAMAN MICROSCOPY

Sample name: Graphene
Ordering customer: Dr. Torsten Busch
Dept./Plant: Curia Germany GmbH
Order number: 24-047052

1. Sample

Sample name	External sample ID
24-047052-1	TB1133
24-047052-2	900561-500MG
24-047052-3	MB2946X_gem
24-047052-4	MB2946A

Table 1: Sample name and external ID

2. General implementation of Raman Microscope

The sample was measured untreated on a quartz glass slide. The performance of Raman Microscope measurements is shown using sample 24-047052-4 (Figure 1). At least 5 dots were defined for the raman spectra. The Raman spectra are normalized to the G band, and the results are averaged. A double determination (two image acquisitions with spectra) was performed in each case. The interpretation of the bands is summarized in Table 1.

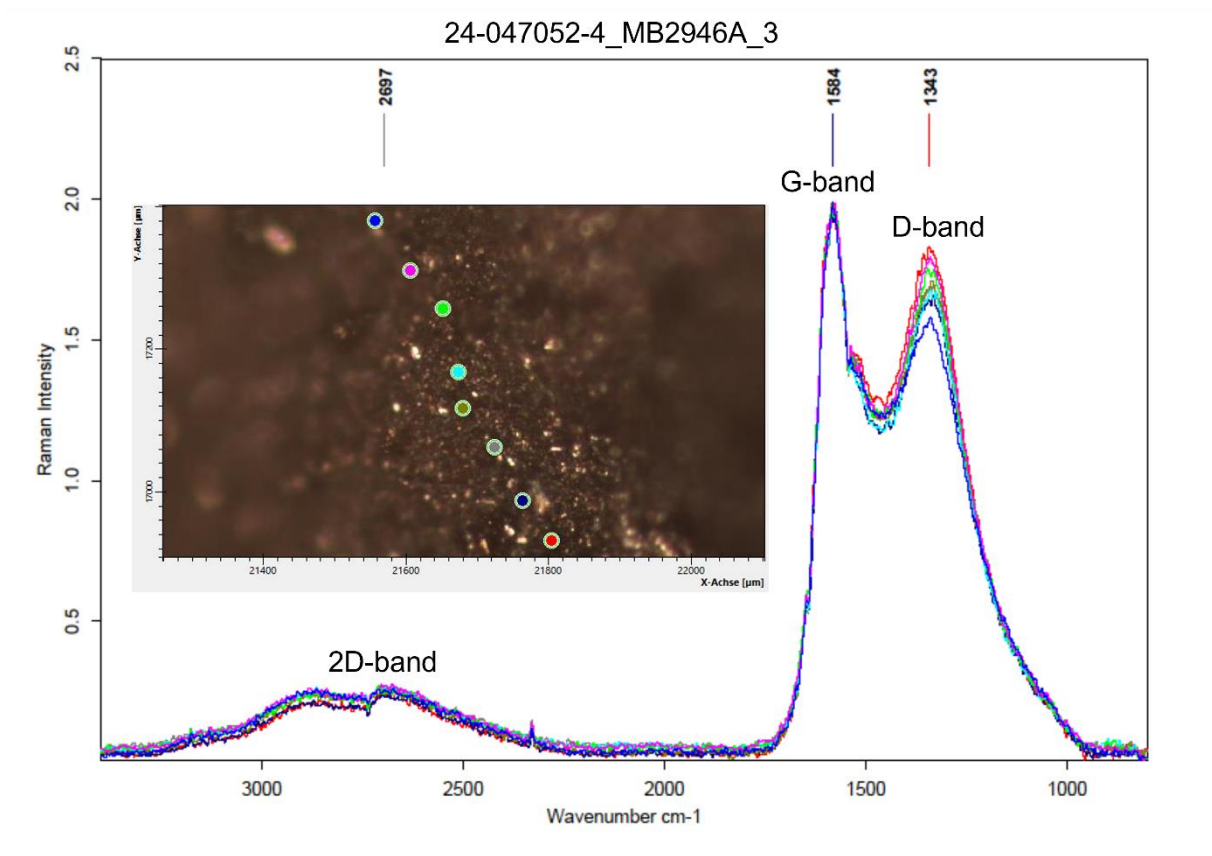


Figure 1: Raman microscopy of sample 24-047052-4. The dots indicate the locations of Raman measurements.

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band	
D	The D-band at around 1350 cm⁻¹ in the Raman spectrum of graphene provides information about defects and disorder in the graphene: - The intensity of the D-band relative to the G-band provides information about the defect density in graphene. The higher the intensity ratio D/G, the higher the defect concentration. - The coupling of graphene to the substrate influences the D-band. A weak coupling is associated with a narrow D-band.
G	This band at around 1580 cm⁻¹ is the fundamental vibration mode of the sp²-hybridized carbon atoms in the graphene plane. Its position depends on the charge carrier density and type, as well as strain in the graphene: - The intensity and shape of the G-band provides information about the number of graphene layers. For monolayer graphene, the band is narrow and sharp, and with increasing number of layers, it broadens and its intensity increases.
2D	This band at around 2680 cm⁻¹ is the overtone of the G-band. Its position is also influenced by the carrier density and type as well as by strain: - The shape of the 2D-band changes with increasing number of layers. For monolayer graphene, it appears as a sharp, symmetric single line. From two layers onwards, it broadens, and a shoulder forms.

Table 2: Explanation of Raman bands
[1] Fromm et al. Contribution of the buffer layer to the Raman spectrum of epitaxial graphene on SiC(0001), New Journal of Physics 15, 043031 (2013).

3. Results

The Raman spectra and the associated images are shown in Figures 2-5.

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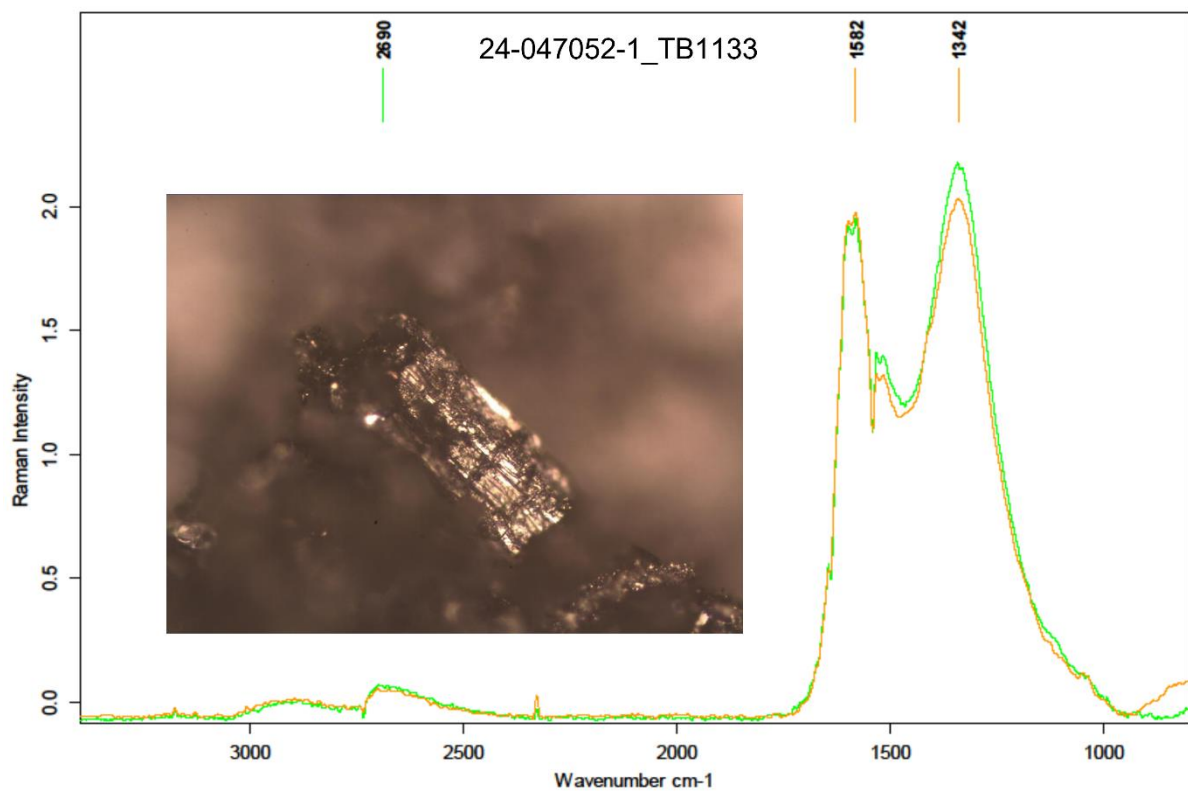


Figure 2: Raman spectra of TB1133 (24-047052-1)

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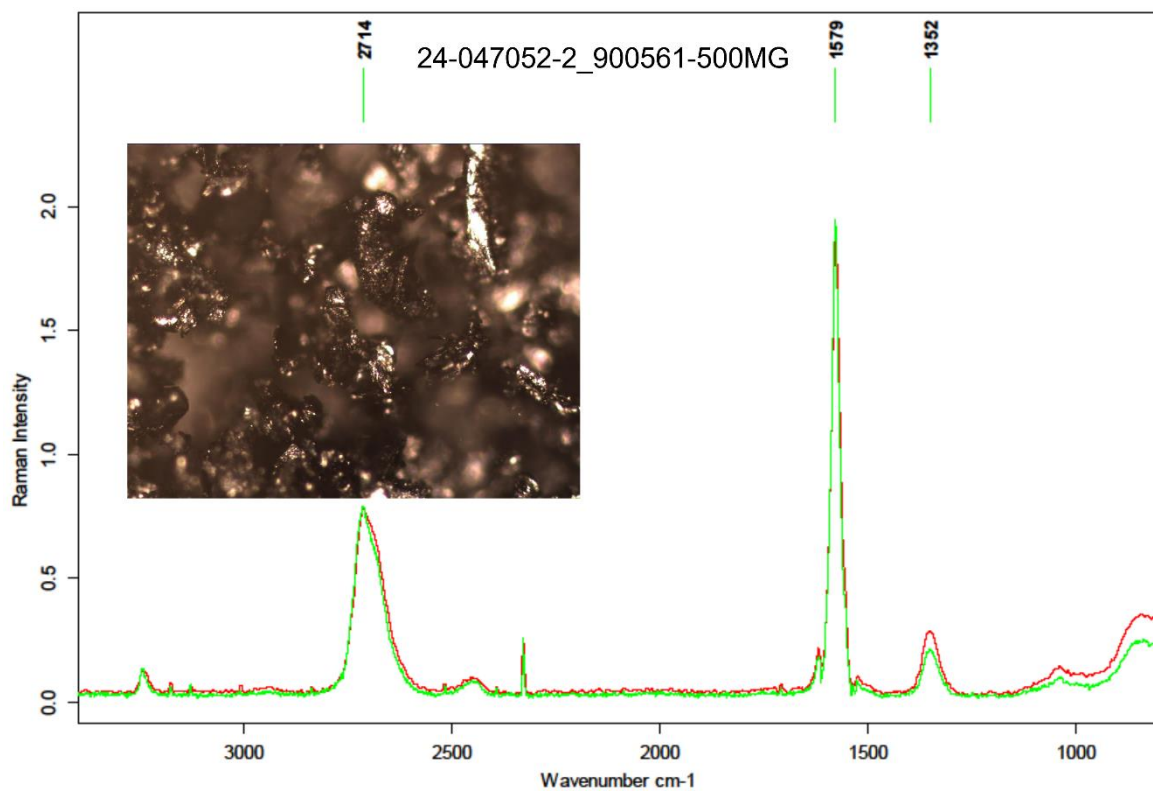


Figure 3: Raman spectra of 900561-500MG (24-047052-2)

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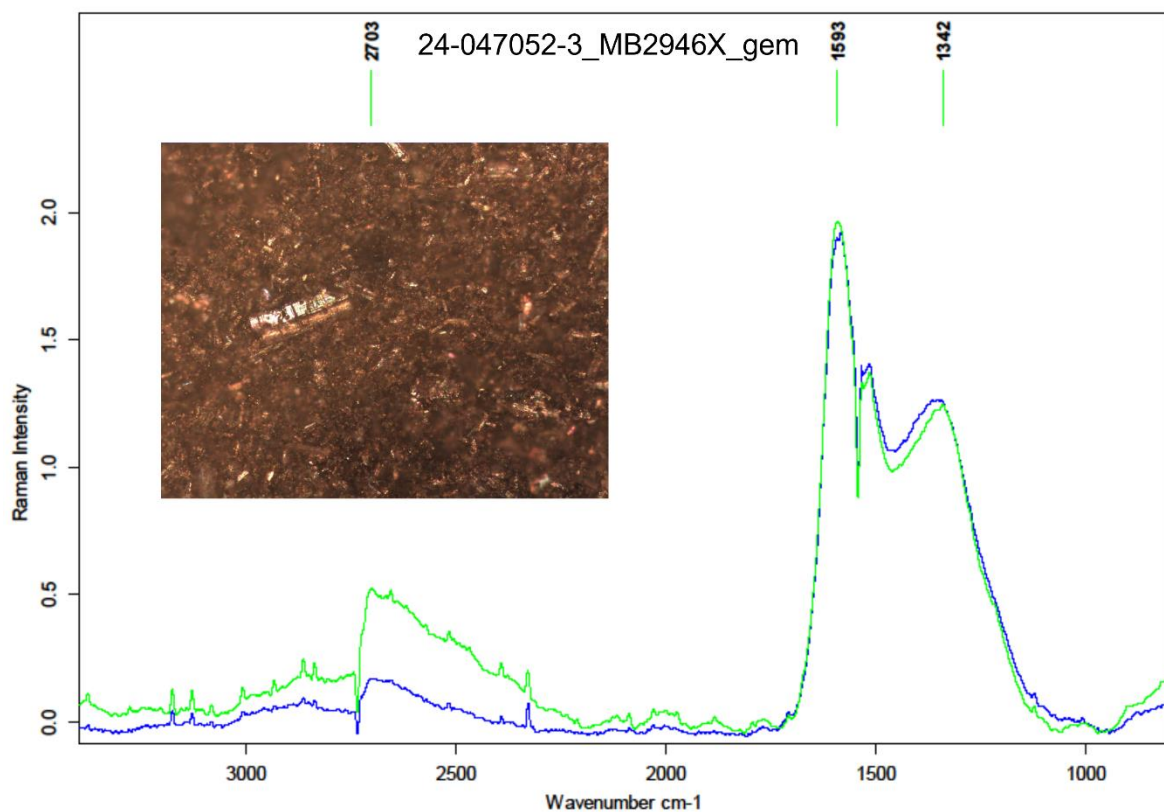


Figure 4: Raman spectra of MB2946X_gem (24-047052-3)

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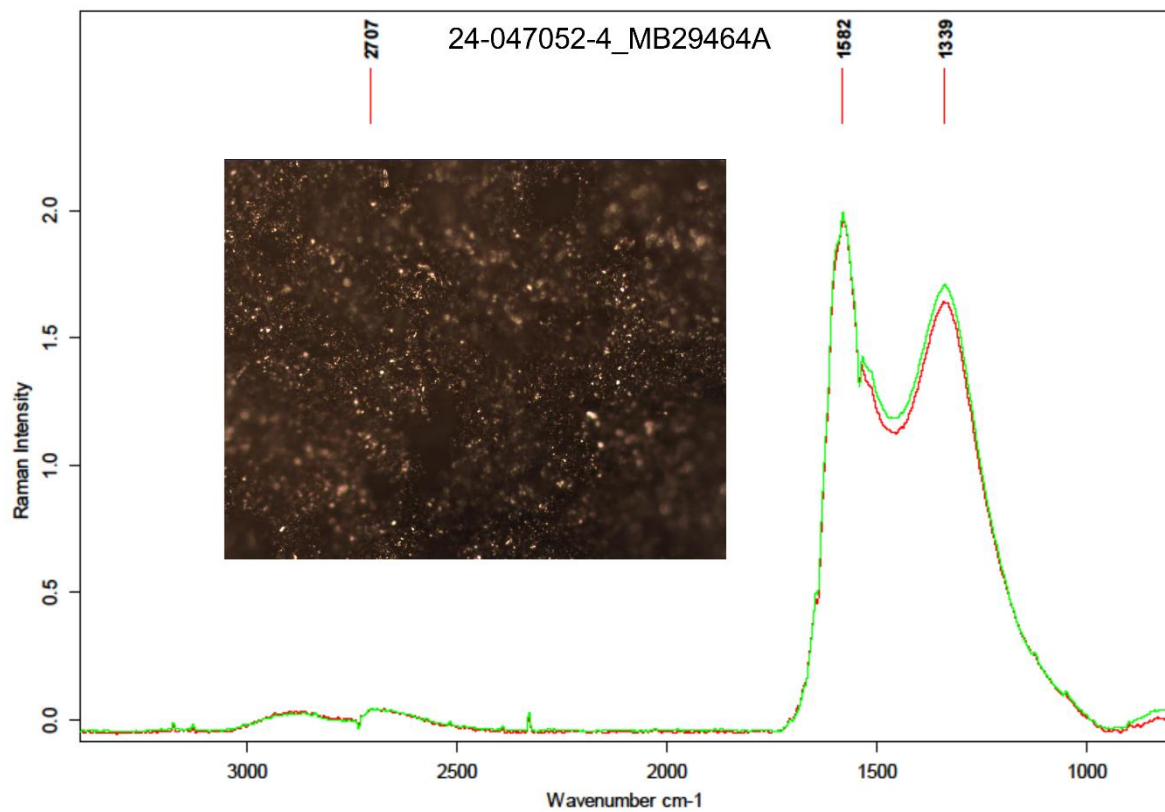


Figure 5: Raman spectra of MB2946A (24-047052-4)